

ON ALMOST PRIMES IN PIATETSKI–SHAPIRO SEQUENCES

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ABSTRACT. The author proves that for $0.992 < \gamma < 1$, there exist infinitely many primes p such that $[p^{1/\gamma}]$ has at most 6 prime factors counted with multiplicity. This gives an improvement upon the previous results of Banks–Guo–Shparlinski and Xue–Li–Zhang.

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1. INTRODUCTION

Let P_r denotes an integer with at most r prime factors counted with multiplicity, p denotes a prime and $\gamma \in (0, 1)$. The Piatetski–Shapiro sequences are sequences of the form

$$\mathcal{N}_\gamma = \left\{ [n^{1/\gamma}] : n \in \mathbb{N}^+ \right\}. \quad (1)$$

In 1953, Piatetski–Shapiro [6] established that there are infinitely many primes in $\mathcal{N}_\gamma$ if $\gamma \in (\frac{11}{12}, 1)$. This range of γ has been improved by many authors, and the best record now is due to Li [5], where he showed that the same result holds for any $\gamma \in (\frac{775}{919}, 1)$.

In 1987, Balog [1] considered the following subsequence of $\mathcal{N}_\gamma$:

$$\mathcal{P}_\gamma = \left\{ [p^{1/\gamma}] : p \text{ prime} \right\}. \quad (2)$$

He showed that for almost all $\gamma \in (0, 1)$, we have

$$\limsup_{x \rightarrow \infty} \frac{\sum_{\substack{q \leq x \\ q \in \mathcal{P}_\gamma}} 1}{x^\gamma \gamma^{-1} (\log x)^{-2}} \geq 1, \quad (3)$$

but this result gives no information for any specific choice of γ .

In 2016, Banks, Guo and Shparlinski [2] generalized and enhanced Balog’s result. They showed explicitly that for every $\gamma \in (0, 1) \setminus \{\frac{1}{z} : z \in \mathbb{Z}^+\}$, there exist infinitely many $q \in \mathcal{P}_\gamma$ such that $q = P_{R(\gamma)}$ for some finite $R(\gamma)$. As a part of their result, they showed that there are infinitely many $q \in \mathcal{P}_\gamma$ with $q = P_8$ for any $\gamma \in (0.9505, 1)$. In 2024, Xue, Li and Zhang [8] improved this to $q = P_7$ for any $\gamma \in (0.989, 1)$. In this paper, we shall further improve this to $q = P_6$ when γ is near 1.

Theorem 1.1. *There are infinitely many $q \in \mathcal{P}_\gamma$ with $q = P_6$ for any $\gamma \in (0.992, 1)$. Moreover, the following estimates*

$$\sum_{\substack{q \leq x \\ q \in \mathcal{P}_\gamma \\ \Omega(q) \leq 6}} 1 \gg \frac{x^\gamma}{(\log x)^2}$$

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holds for all sufficiently large x provided that $\gamma \in (0.992, 1)$.

Throughout this paper, we always suppose that x is a sufficiently large integer, ε and η are sufficiently small positive numbers and $0 < \gamma < 1$. Let C_0 denote Euler's constant. The letter p , with or without subscript, is reserved for prime numbers.

2. PRELIMINARY LEMMAS

Let $\mathcal{A}$ denote a finite set of positive integers and $z \geq 2$. For square-free d , put

$$P(z) = \prod_{p < z} p, \quad \mathcal{A}_d = \{a : a \in \mathcal{A}, d \mid a\}, \quad S(\mathcal{A}, z) = \sum_{\substack{a \in \mathcal{A} \\ (a, P(z))=1}} 1.$$

The first lemma is a classical result of the linear sieve.

Lemma 2.1. ([4], Pages 205–209). Suppose that every $|\mathcal{A}_d|$ can be written as

$$|\mathcal{A}_d| = \frac{\omega(d)}{d} X_{\mathcal{A}} + r(X_{\mathcal{A}}, d),$$

where $\omega(d)$ is a multiplicative function such that $0 \leq \omega(p) < p$, and $X_{\mathcal{A}} > 1$ is independent of d . Assume further that

$$\sum_{z_1 \leq p < z_2} \frac{\omega(p)}{p} = \log \frac{\log z_2}{\log z_1} + O\left(\frac{1}{\log z_1}\right), \quad z_2 > z_1 \geq 2.$$

Then

$$\begin{aligned} S(\mathcal{A}, z) &\geq X_{\mathcal{A}} W(z) \left\{ f\left(\frac{\log D}{\log z}\right) + O\left(\frac{1}{\log^{\frac{1}{3}} D}\right) \right\} - \sum_{\substack{d \leq D \\ d \mid P(z)}} |r(X_{\mathcal{A}}, d)|, \\ S(\mathcal{A}, z) &\leq X_{\mathcal{A}} W(z) \left\{ F\left(\frac{\log D}{\log z}\right) + O\left(\frac{1}{\log^{\frac{1}{3}} D}\right) \right\} + \sum_{\substack{d \leq D \\ d \mid P(z)}} |r(X_{\mathcal{A}}, d)|, \end{aligned}$$

where D is a power of z ,

$$W(z) = \prod_{p \mid P(z)} \left(1 - \frac{\omega(p)}{p}\right),$$

and $f(s)$ and $F(s)$ are determined by the following differential-difference equation

$$\begin{cases} F(s) = \frac{2 \exp(C_0)}{s}, & f(s) = 0, & 0 < s \leq 2, \\ (sF(s))' = f(s-1), & (sf(s))' = F(s-1), & s \geq 2. \end{cases}$$

Next, we shall prove two combinatorial lemmas, which help us deal with the sums of numbers with 7 or 8 prime factors in the next section.

Lemma 2.2. For any primes $p_1, p_2, p_3, p_4, p_5, p_6, p_7, p_8$ such that

$$x^{\frac{1}{17.41}} \leq p_1 \leq p_2 \leq p_3 \leq p_4 \leq p_5 \leq p_6 \leq p_7 \leq p_8 \quad \text{and} \quad x^{0.999} < p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 \leq x,$$

There exists some $I \subset \{1, 2, 3, 4, 5, 6, 7, 8\}$ such that

$$x^{0.631} < \prod_{i \in I} p_i < x^{0.784}.$$

Proof. Since $p_5 p_6 p_7 p_8 \geq p_1 p_2 p_3 p_4$ and $x^{0.999} < p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8$, we have

$$p_5 p_6 p_7 p_8 \geq (p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8)^{\frac{1}{2}} > x^{0.499}.$$

Assume that $x^{0.631} < p_5 p_6 p_7 p_8 < x^{0.784}$. Now we can take $I = \{5, 6, 7, 8\}$.

Assume that $x^{0.784} \leq p_5 p_6 p_7 p_8$. Since we have $x^{\frac{1}{17.41}} \leq p_1 \leq p_2 \leq p_3 \leq p_4$, we get

$$x \geq p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 \geq x^{0.784 + \frac{4}{17.41}} > x^{1.013} > x,$$

making a contradiction.

Assume that $x^{0.499} < p_5 p_6 p_7 p_8 \leq x^{0.631}$. Now we must have $x^{0.368} \leq p_1 p_2 p_3 p_4 \leq x^{0.501}$, since otherwise we get

$$p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8 \leq x^{0.999} \quad \text{or} \quad x < p_1 p_2 p_3 p_4 p_5 p_6 p_7 p_8,$$

making a contradiction. Since $p_1 \leq p_2 \leq p_3 \leq p_4$, we have $p_1 \leq (p_1 p_2 p_3 p_4)^{\frac{1}{4}} < x^{0.126}$. Now we have

$$p_1 p_5 p_6 p_7 p_8 \leq x^{0.126+0.631} = x^{0.757} < x^{0.784}.$$

If we have $p_1 p_5 p_6 p_7 p_8 > x^{0.631}$, we can take $I = \{1, 5, 6, 7, 8\}$. Otherwise we have $x^{0.499+\frac{1}{17.41}} < p_1 p_5 p_6 p_7 p_8 \leq x^{0.631}$, and thus $x^{0.368} \leq p_2 p_3 p_4 \leq x^{0.501-\frac{1}{17.41}}$. Since $p_2 \leq (p_2 p_3 p_4)^{\frac{1}{3}} < x^{0.148}$, we have

$$x^{0.613} < x^{0.499+\frac{2}{17.41}} < p_1 p_2 p_5 p_6 p_7 p_8 < x^{0.631+0.148} = x^{0.779}.$$

If we have $p_1 p_2 p_5 p_6 p_7 p_8 > x^{0.631}$, we can take $I = \{1, 2, 5, 6, 7, 8\}$. Otherwise we have $x^{0.613} < p_1 p_2 p_5 p_6 p_7 p_8 \leq x^{0.631}$, and thus $x^{0.368} \leq p_3 p_4 \leq x^{0.387}$. But now we get

$$p_5 p_6 p_7 p_8 \geq (p_3 p_4)^2 \geq x^{2 \cdot 0.368} = x^{0.736},$$

making a contradiction with the assumption $p_5 p_6 p_7 p_8 \leq x^{0.631}$.

Combining all cases above, Lemma 2.2 is proved. $\square$

Lemma 2.3. *For any primes $p_1, p_2, p_3, p_4, p_5, p_6, p_7$ such that*

$$x^{\frac{1}{17.41}} \leq p_1 \leq p_2 \leq p_3 \leq p_4 \leq p_5 \leq p_6 \leq p_7 \quad \text{and} \quad x^{0.999} < p_1 p_2 p_3 p_4 p_5 p_6 p_7 \leq x,$$

There exists some $I \subset \{1, 2, 3, 4, 5, 6, 7\}$ such that

$$x^{0.631} < \prod_{i \in I} p_i < x^{0.784}.$$

Proof. Assume that $x^{0.631} < p_4 p_5 p_6 p_7 < x^{0.784}$. Now we can take $I = \{4, 5, 6, 7\}$.

Assume that $x^{0.784} \leq p_4 p_5 p_6 p_7$. Now we have $p_1 p_2 p_3 \leq x^{0.216}$. Since $x^{\frac{1}{17.41}} \leq p_1 \leq p_2 \leq p_3$, we have

$$p_4 p_5 p_6 p_7 = \frac{p_1 p_2 p_3 p_4 p_5 p_6 p_7}{p_1 p_2 p_3} \leq x^{1-\frac{3}{17.41}} < x^{0.828}.$$

Now

$$x^{\frac{1}{17.41}} \leq p_4 \leq (p_4 p_5 p_6 p_7)^{\frac{1}{4}} < x^{0.207},$$

we have

$$x^{0.576} < x^{0.784-0.207} \leq \frac{p_4 p_5 p_6 p_7}{p_4} = p_5 p_6 p_7 = \frac{p_4 p_5 p_6 p_7}{p_4} \leq x^{0.828-\frac{1}{17.41}} < x^{0.771}.$$

If we have $x^{0.631} < p_5 p_6 p_7 < x^{0.771}$, we can take $I = \{5, 6, 7\}$. Otherwise we have $x^{0.576} < p_5 p_6 p_7 \leq x^{0.631}$. Since

$$x^{\frac{1}{17.41}} \leq p_1 \leq (p_1 p_2 p_3 p_4 p_5 p_6 p_7)^{\frac{1}{7}} < x^{0.143},$$

we have

$$x^{0.633} < x^{0.576+\frac{1}{17.41}} \leq p_1 p_5 p_6 p_7 \leq x^{0.631+0.143} < x^{0.774},$$

and we can take $I = \{1, 5, 6, 7\}$.

Assume that $p_4 p_5 p_6 p_7 \leq x^{0.631}$. Now we must have $p_1 p_2 p_3 \geq x^{0.368}$, and thus $p_3 \geq (p_1 p_2 p_3)^{\frac{1}{3}} > x^{0.122}$. Now we have $x^{0.122} < p_3 \leq p_4 \leq p_5 \leq p_6 \leq p_7$. Note that

$$p_1 p_2 p_3 \leq (p_1 p_2 p_3 p_4 p_5 p_6)^{\frac{1}{2}} \leq x^{\frac{3}{7}}$$

and

$$p_4 p_5 \leq (p_4 p_5 p_6 p_7)^{\frac{1}{2}} \leq x^{\frac{0.631}{2}} < x^{0.316},$$

we have

$$x^{0.612} = x^{0.368+2 \cdot 0.122} < p_1 p_2 p_3 p_4 p_5 < x^{\frac{3}{7}+0.316} < x^{0.745}.$$

If we have $p_1 p_2 p_3 p_4 p_5 > x^{0.631}$, we can take $I = \{1, 2, 3, 4, 5\}$. Otherwise we have $x^{0.612} < p_1 p_2 p_3 p_4 p_5 \leq x^{0.631}$, and thus $x^{0.368} \leq p_6 p_7 \leq x^{0.388}$. Now we have $p_7 \geq x^{0.184}$, and thus

$$p_3 p_4 p_5 p_6 p_7 \geq x^{0.184+4 \cdot 0.122} > x^{0.67}.$$

Since we also have

$$p_3 p_4 p_5 p_6 p_7 \leq (p_4 p_5)^{\frac{1}{2}} p_4 p_5 p_6 p_7 < x^{\frac{0.316}{2}+0.631} = x^{0.789},$$

if we have $p_3 p_4 p_5 p_6 p_7 < x^{0.784}$, we can take $I = \{3, 4, 5, 6, 7\}$. Otherwise we have $x^{0.784} \leq p_3 p_4 p_5 p_6 p_7 < x^{0.789}$, and thus $x^{0.21} \leq p_1 p_2 \leq x^{0.216}$. Now we have

$$x^{0.7} = x^{0.21+0.122+0.368} \leq p_1 p_2 p_3 p_6 p_7 \leq p_1 p_2 (p_4 p_5)^{\frac{1}{2}} p_6 p_7 < x^{0.216+\frac{0.316}{2}+0.388} = x^{0.762},$$

and we can take $I = \{1, 2, 3, 6, 7\}$.

Combining all cases above, Lemma 2.3 is proved. $\square$

3. PROOF OF THEOREM 1.1

We define the set $\mathcal{A}$ as

$$\mathcal{A} = \{a : a \leq x, a \in \mathcal{P}_\gamma\}.$$

Let $\xi = \frac{140\gamma-99}{270} - \eta$, $u = \frac{1}{\xi} + \varepsilon$ and $\lambda = \frac{1}{9-u-\varepsilon}$ as in [8]. We shall prove our Theorem 1.1 using Richert's weighted sieve [7] together with Chen's switching principle [3]. Consider the following weighted sum

$$W\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) = \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1}} \left(1 - \lambda \sum_{\substack{x^{\frac{1}{17.41}} \leq p < x^{\frac{1}{u}} \\ p|a}} \left(1 - \frac{u \log p}{\log x}\right)\right) = \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1}} \mathcal{W}_a. \quad (4)$$

We have

$$\begin{aligned} W\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) &= \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1}} \mathcal{W}_a \\ &= \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a) \leq 6}} \mathcal{W}_a + \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=8 \\ \mu(a) \neq 0}} \mathcal{W}_a + \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=7 \\ \mu(a) \neq 0}} \mathcal{W}_a \\ &\quad + \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ 7 \leq \Omega(a) \leq 8 \\ \mu(a)=0}} \mathcal{W}_a + \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a) \geq 9}} \mathcal{W}_a \\ &= W_1 + W_2 + W_3 + W_4 + W_5. \end{aligned} \quad (5)$$

Our aim is to show that $W_1 > 0$.

Trivially, we have

$$W_4 = \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ 7 \leq \Omega(a) \leq 8 \\ \mu(a)=0}} \mathcal{W}_a \ll \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \mu(a)=0}} \tau(a) \ll x^\varepsilon \sum_{x^{\frac{1}{17.41}} \leq p \leq x^{\frac{1}{2}}} \left(\frac{x}{p^2} + 1\right) \ll x^{1-\frac{1}{17.41}+\varepsilon}.$$

Since

$$\begin{aligned} \mathcal{W}_a &= 1 - \lambda \sum_{\substack{x^{\frac{1}{17.41}} \leq p < x^{\frac{1}{u}} \\ p|a}} \left(1 - \frac{u \log p}{\log x}\right) \leq \lambda \left(\frac{1}{\lambda} - \sum_{p|a} \left(1 - \frac{u \log p}{\log x}\right)\right) \\ &= \lambda \left(9 - u - \varepsilon - \Omega(a) + \frac{u \log a}{\log x}\right) < \lambda(9 - \Omega(a)), \end{aligned} \quad (6)$$

we have $\mathcal{W}_a < 0$ for $\Omega(a) \geq 9$. Thus,

$$\begin{aligned} W_1 &= W\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) - W_2 - W_3 - W_4 - W_5 \\ &= W\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) - W_2 - W_3 - W_5 + O\left(x^{1-10^{-10}}\right) \end{aligned}$$

$$\begin{aligned}
&> W\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) - W_2 - W_3 + O\left(x^{1-10^{-10}}\right) \\
&= S\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) - \lambda \sum_{x^{\frac{1}{17.41}} \leq p < x^{\frac{1}{u}}} \left(1 - \frac{u \log p}{\log x}\right) S\left(\mathcal{A}_p, x^{\frac{1}{17.41}}\right) - W_2 - W_3 + O\left(x^{1-10^{-10}}\right) \\
&= S_1 - S_2 - W_2 - W_3 + O\left(x^{1-10^{-10}}\right).
\end{aligned} \tag{7}$$

We first give a lower bound for S_1 . Taking

$$X_{\mathcal{A}} = \pi(x^\gamma), \quad \omega(d) = \mathbb{1}_{d|P}\left(x^{\frac{1}{17.41}}\right)$$

in Lemma 2.1. By [[7], Lemma 2] we know that

$$W(z) = \mathcal{C}(\omega) \frac{\exp(-C_0)}{\log z} \left(1 + O\left(\frac{1}{\log z}\right)\right), \quad \text{where } \mathcal{C}(\omega) = \prod_p \left(1 - \frac{\omega(p)}{p}\right) \left(1 - \frac{1}{p}\right)^{-1}. \tag{8}$$

From the arguments in [[8], Section 4], we know that

$$\sum_{d \leq x^\xi} |r(X_{\mathcal{A}}, d)| = \sum_{d \leq x^\xi} \left| |\mathcal{A}_d| - \frac{\omega(d)}{d} X_{\mathcal{A}} \right| \ll \frac{x^\gamma}{(\log x)^A}. \tag{9}$$

Thus, we can take $D = x^\xi$ in Lemma 2.1. By Lemma 2.1, (8) and the fact that $2 < 17.41\xi < 4$ holds for $0.93 < \gamma < 1$, we have

$$\begin{aligned}
S_1 &= S\left(\mathcal{A}, x^{\frac{1}{17.41}}\right) \geq (1 + o(1)) \pi(x^\gamma) W\left(x^{\frac{1}{17.41}}\right) f(17.41\xi) \\
&= (1 + o(1)) 2 \exp(C_0) \pi(x^\gamma) W\left(x^{\frac{1}{17.41}}\right) \left(\frac{\log(17.41\xi - 1)}{17.41\xi}\right) \\
&= (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{\log(17.41\xi - 1)}{\xi}\right).
\end{aligned} \tag{10}$$

Similarly, we can give an upper bound for S_2 :

$$\begin{aligned}
S_2 &= \lambda \sum_{x^{\frac{1}{17.41}} \leq p < x^{\frac{1}{u}}} \left(1 - \frac{u \log p}{\log x}\right) S\left(\mathcal{A}_p, x^{\frac{1}{17.41}}\right) \\
&\leq (1 + o(1)) \pi(x^\gamma) W\left(x^{\frac{1}{17.41}}\right) \left(\lambda \sum_{x^{\frac{1}{17.41}} \leq p < x^{\frac{1}{u}}} \left(1 - \frac{u \log p}{\log x}\right) \frac{F\left(\frac{\log\left(\frac{x^\xi}{p}\right)}{\log\left(x^{\frac{1}{17.41}}\right)}\right)}{p} \right) \\
&= (1 + o(1)) 2 \exp(C_0) \pi(x^\gamma) W\left(x^{\frac{1}{17.41}}\right) \left(\lambda \int_u^{17.41} \frac{t - u}{17.41t(t\xi - 1)} dt \right) \\
&= (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\lambda \int_u^{17.41} \frac{t - u}{t(t\xi - 1)} dt \right).
\end{aligned} \tag{11}$$

Next, we give an upper bound for W_2 . By (6), we have

$$\begin{aligned}
W_2 &= \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=8 \\ \mu(a) \neq 0}} \mathcal{W}_a < \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=8 \\ \mu(a) \neq 0}} \lambda(9 - \Omega(a)) \\
&= \lambda \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=8 \\ \mu(a) \neq 0}} 1 = \lambda \sum_{\substack{[p^{1/\gamma}] \leq x \\ [p^{1/\gamma}] = p_1 \cdots p_8 \\ x^{\frac{1}{17.41}} \leq p_1 < \cdots < p_8}} 1.
\end{aligned} \tag{12}$$

Define

$$\mathcal{B}^1 = \left\{ n : n \leq x, n = p_1 \cdots p_8, x^{\frac{1}{17.41}} \leq p_1 < \cdots < p_8 \right\} \quad \text{and} \quad \mathcal{E}^1 = \left\{ a : [a^{1/\gamma}] \in \mathcal{B}^1 \right\}. \quad (13)$$

Then by (12) and (13) we know that

$$W_2 = \lambda S \left(\mathcal{E}^1, x^{\frac{\gamma}{2}} \right). \quad (14)$$

Write $[(dm)^{1/\gamma}] = l \in \mathcal{B}^1$, we have

$$\begin{aligned} |\mathcal{E}_d^1| &= \sum_{\substack{m \\ [(dm)^{1/\gamma}] \in \mathcal{B}^1}} 1 = \sum_{l \in \mathcal{B}^1} \left(\left[-\frac{l^\gamma}{d} \right] - \left[-\frac{(l+1)^\gamma}{d} \right] \right) \\ &= \frac{1}{d} \sum_{l \in \mathcal{B}^1} ((l+1)^\gamma - l^\gamma) + \sum_{l \in \mathcal{B}^1} \left(\psi \left(-\frac{(l+1)^\gamma}{d} \right) - \psi \left(-\frac{l^\gamma}{d} \right) \right) \\ &= \frac{1}{d} X_{\mathcal{B}^1} + r(X_{\mathcal{B}^1}, d), \end{aligned} \quad (15)$$

where $\psi(t) = t - [t] - \frac{1}{2}$ is the sawtooth function. By the proof of [[8], Lemma 5.1], we know that we have

$$\sum_{d \leq x^\xi} |r(X_{\mathcal{B}^1}, d)| \ll \frac{x^\gamma}{(\log x)^A} \quad (16)$$

under the condition that a partial product of $p_1 \cdots p_8$ lies in the interval $[X^{5-5\gamma+4\xi+\eta}, X^{\frac{1}{4}(\gamma+\xi+2)-\eta}]$, where $x^{1-\eta} < X \leq x$. Since we have

$$5 - 5\gamma + 4\xi < 0.6309 \quad \text{and} \quad \frac{1}{4}(\gamma + \xi + 2) > 0.7849$$

when $0.992 < \gamma < 1$, this condition is satisfied by Lemma 2.2. Now, using Lemma 2.1 and by (8), we get

$$\begin{aligned} W_2 &= \lambda S \left(\mathcal{E}^1, x^{\frac{\gamma}{2}} \right) \leq \lambda(1 + o(1)) X_{\mathcal{B}^1} W \left(x^{\frac{\gamma}{2}} \right) F \left(\frac{2\xi}{\gamma} \right) \\ &= (1 + o(1)) X_{\mathcal{B}^1} \frac{2\mathcal{C}(\omega)\lambda}{17.41^2 \xi \log x}. \end{aligned} \quad (17)$$

Using the Prime Number Theorem, we have

$$\begin{aligned} X_{\mathcal{B}^1} &= (1 + o(1)) \pi(x^\gamma) \left(\gamma \int_{\frac{1}{17.41}}^{\frac{1}{8}} \int_{t_1}^{\frac{1}{7}(1-t_1)} \int_{t_2}^{\frac{1}{6}(1-t_1-t_2)} \int_{t_3}^{\frac{1}{5}(1-t_1-t_2-t_3)} \int_{t_4}^{\frac{1}{4}(1-t_1-t_2-t_3-t_4)} \right. \\ &\quad \left. \int_{t_5}^{\frac{1}{3}(1-t_1-t_2-t_3-t_4-t_5)} \int_{t_6}^{\frac{1}{2}(1-t_1-t_2-t_3-t_4-t_5-t_6)} \right. \\ &\quad \left. \frac{1}{t_1 t_2 t_3 t_4 t_5 t_6 t_7 (1-t_1-t_2-t_3-t_4-t_5-t_6-t_7)} dt_7 dt_6 dt_5 dt_4 dt_3 dt_2 dt_1 \right). \end{aligned} \quad (18)$$

Combining (17) and (18), we obtain

$$\begin{aligned} W_2 &\leq (1 + o(1)) X_{\mathcal{B}^1} \frac{2\mathcal{C}(\omega)\lambda}{17.41^2 \xi \log x} \\ &= (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{\lambda\gamma}{\xi} \int_{\frac{1}{17.41}}^{\frac{1}{8}} \int_{t_1}^{\frac{1}{7}(1-t_1)} \int_{t_2}^{\frac{1}{6}(1-t_1-t_2)} \int_{t_3}^{\frac{1}{5}(1-t_1-t_2-t_3)} \int_{t_4}^{\frac{1}{4}(1-t_1-t_2-t_3-t_4)} \right. \\ &\quad \left. \int_{t_5}^{\frac{1}{3}(1-t_1-t_2-t_3-t_4-t_5)} \int_{t_6}^{\frac{1}{2}(1-t_1-t_2-t_3-t_4-t_5-t_6)} \right. \\ &\quad \left. \frac{1}{t_1 t_2 t_3 t_4 t_5 t_6 t_7 (1-t_1-t_2-t_3-t_4-t_5-t_6-t_7)} dt_7 dt_6 dt_5 dt_4 dt_3 dt_2 dt_1 \right) \end{aligned}$$

$$\leq (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{\lambda\gamma}{\xi} \cdot 0.003 \right). \quad (19)$$

Similarly, we can give an upper bound for W_3 . By (6), we have

$$\begin{aligned} W_3 &= \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=7 \\ \mu(a) \neq 0}} \mathcal{W}_a < \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=7 \\ \mu(a) \neq 0}} \lambda(9 - \Omega(a)) \\ &= \lambda \sum_{\substack{a \in \mathcal{A} \\ (a, P(x^{\frac{1}{17.41}}))=1 \\ \Omega(a)=7 \\ \mu(a) \neq 0}} 2 = 2\lambda \sum_{\substack{[p^{1/\gamma}] \leq x \\ [p^{1/\gamma}] = p_1 \cdots p_7 \\ x^{\frac{1}{17.41}} \leq p_1 < \cdots < p_7}} 1. \end{aligned} \quad (20)$$

Define

$$\mathcal{B}^2 = \left\{ n : n \leq x, n = p_1 \cdots p_7, x^{\frac{1}{17.41}} \leq p_1 < \cdots < p_7 \right\} \quad \text{and} \quad \mathcal{E}^2 = \left\{ a : [a^{1/\gamma}] \in \mathcal{B}^2 \right\}. \quad (21)$$

Then by (20) and (21) we know that

$$W_3 = 2\lambda S\left(\mathcal{E}^2, x^{\frac{\gamma}{2}}\right). \quad (22)$$

Write $[(dm)^{1/\gamma}] = l \in \mathcal{B}^1$, we have

$$\begin{aligned} |\mathcal{E}_d^2| &= \sum_{\substack{m \\ [(dm)^{1/\gamma}] \in \mathcal{B}^2}} 1 = \sum_{l \in \mathcal{B}^2} \left(\left[-\frac{l^\gamma}{d} \right] - \left[-\frac{(l+1)^\gamma}{d} \right] \right) \\ &= \frac{1}{d} \sum_{l \in \mathcal{B}^2} ((l+1)^\gamma - l^\gamma) + \sum_{l \in \mathcal{B}^2} \left(\psi\left(-\frac{(l+1)^\gamma}{d}\right) - \psi\left(-\frac{l^\gamma}{d}\right) \right) \\ &= \frac{1}{d} X_{\mathcal{B}^2} + r(X_{\mathcal{B}^2}, d), \end{aligned} \quad (23)$$

where $\psi(t) = t - [t] - \frac{1}{2}$ is the sawtooth function. By the proof of [[8], Lemma 5.1], we know that we have

$$\sum_{d \leq x^\xi} |r(X_{\mathcal{B}^2}, d)| \ll \frac{x^\gamma}{(\log x)^A} \quad (24)$$

under the condition that a partial product of $p_1 \cdots p_8$ lies in the interval $\left[X^{5-5\gamma+4\xi+\eta}, X^{\frac{1}{4}(\gamma+\xi+2)-\eta} \right]$, where $x^{1-\eta} < X \leq x$. Since we have

$$5 - 5\gamma + 4\xi < 0.6309 \quad \text{and} \quad \frac{1}{4}(\gamma + \xi + 2) > 0.7849$$

when $0.992 < \gamma < 1$, this condition is satisfied by Lemma 2.3. Now, using Lemma 2.1 and by (8), we get

$$\begin{aligned} W_3 &= 2\lambda S\left(\mathcal{E}^2, x^{\frac{\gamma}{2}}\right) \leq 2\lambda(1 + o(1)) X_{\mathcal{B}^2} W\left(x^{\frac{\gamma}{2}}\right) F\left(\frac{2\xi}{\gamma}\right) \\ &= (1 + o(1)) X_{\mathcal{B}^2} \frac{4\mathcal{C}(\omega)\lambda}{17.41^2 \xi \log x}. \end{aligned} \quad (25)$$

Using the Prime Number Theorem, we have

$$\begin{aligned} X_{\mathcal{B}^2} &= (1 + o(1)) \pi(x^\gamma) \left(\gamma \int_{\frac{1}{17.41}}^{\frac{1}{7}} \int_{t_1}^{\frac{1}{6}(1-t_1)} \int_{t_2}^{\frac{1}{5}(1-t_1-t_2)} \int_{t_3}^{\frac{1}{4}(1-t_1-t_2-t_3)} \right. \\ &\quad \left. \int_{t_4}^{\frac{1}{3}(1-t_1-t_2-t_3-t_4)} \int_{t_5}^{\frac{1}{2}(1-t_1-t_2-t_3-t_4-t_5)} \right. \\ &\quad \left. \frac{1}{t_1 t_2 t_3 t_4 t_5 t_6 (1-t_1-t_2-t_3-t_4-t_5-t_6)} dt_6 dt_5 dt_4 dt_3 dt_2 dt_1 \right). \end{aligned} \quad (26)$$

Combining (25) and (26), we obtain

$$\begin{aligned}
W_3 &\leq (1 + o(1)) X_{B^2} \frac{4\mathcal{C}(\omega)\lambda}{17.41^2 \xi \log x} \\
&= (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{2\lambda\gamma}{\xi} \int_{\frac{1}{17.41}}^{\frac{1}{7}} \int_{t_1}^{\frac{1}{6}(1-t_1)} \int_{t_2}^{\frac{1}{5}(1-t_1-t_2)} \int_{t_3}^{\frac{1}{4}(1-t_1-t_2-t_3)} \right. \\
&\quad \left. \int_{t_4}^{\frac{1}{3}(1-t_1-t_2-t_3-t_4)} \int_{t_5}^{\frac{1}{2}(1-t_1-t_2-t_3-t_4-t_5)} \frac{1}{t_1 t_2 t_3 t_4 t_5 t_6 (1-t_1-t_2-t_3-t_4-t_5-t_6)} dt_6 dt_5 dt_4 dt_3 dt_2 dt_1 \right) \\
&\leq (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{\lambda\gamma}{\xi} \cdot 0.052 \right). \tag{27}
\end{aligned}$$

Finally, by (10), (11), (19) and (27) we get that

$$\begin{aligned}
W_1 &> S_1 - S_2 - W_2 - W_3 + O\left(x^{1-10^{-10}}\right) \\
&\geq (1 + o(1)) \frac{2\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x} \left(\frac{\log(17.41\xi - 1)}{\xi} - \lambda \int_u^{17.41} \frac{t-u}{t(t\xi - 1)} dt - \frac{\lambda\gamma}{\xi} (0.003 + 0.052) \right). \tag{28}
\end{aligned}$$

For $0.992 < \gamma < 1$, numerical calculations give that

$$\frac{\log(17.41\xi - 1)}{\xi} - \lambda \int_u^{17.41} \frac{t-u}{t(t\xi - 1)} dt - \frac{\lambda\gamma}{\xi} (0.003 + 0.052) > 0.026. \tag{29}$$

Thus, we have

$$W_1 > \frac{0.052\mathcal{C}(\omega)\pi(x^\gamma)}{17.41^2 \log x}, \tag{30}$$

and the proof of Theorem 1.1 is completed.

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